

Avinash Karri

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SUMMARY

AI/ML Software Engineer with hands-on experience building **production-grade ML/LLM inference systems**, developing **end-to-end data pipelines**, and deploying agentic and GenAI applications in cloud environments (Azure & GCP). Skilled in Python, REST API design, RAG systems, LLM orchestration, containerized deployments, and ML workflow automation. Experienced in collaborating across data science, product, and engineering teams to deliver secure, scalable, and business-critical AI solutions.

WORK EXPERIENCE

ALSTOM

Bengaluru

Associate System Application Design Engineer

Jul 2024 - Present

- Eliminated 100% manual intervention in key engineering workflows by developing robust Python-based automation tools.
- Increased documentation speed by 50% with AI-driven utilities using LLMs for automated data extraction and content generation.
- Partnered with cross-functional teams to identify bottlenecks and deliver automation solutions, driving a 25% boost in overall process efficiency.

Fluentgrid Limited

Visakhapatnam

Fullstack Developer

May 2023 - Jun 2023

- Collaborated with software engineers to develop a New Service Connection portal for customers to request electricity connections in cities eliminated the need for physical applications by 100%.
- Developed and implemented a secure and user-friendly portal for customers to request electricity connections in cities.

EDUCATION

GITAM University

Visakhapatnam

Bachelor of Technology, Computer Science Engineering

Graduation Date: Apr 2024

PROJECT EXPERIENCE

Cypher AI Assistant

Agentic Personal Automation

- Built a **LangGraph + Vertex AI** multi-agent orchestration system enabling autonomous task execution across payments, travel, productivity, and developer workflows, with long-term vector memory for contextual personalization.
- Integrated **PayPal API, Google Maps API, GitHub MCP, and Playwright** to automate real-world actions like payments, navigation, trip planning, repository analysis, web bookings, and dynamic page interactions — reducing manual effort for common tasks.
- Designed a modular FastAPI backend with tool plugins (email parsing, expense extraction, scheduling, notifications), enabling complex workflows (e.g., “book trip → auto-pay → share route → log expense”) with **fully automated end-to-end execution**.

Asteria AIOps

Production-Grade ML & GenAI Ops

- Python, FastAPI, TorchServe, Docker, LLMs, ChromaDB, GitHub Actions, Azure/GCP
- Built a full-stack AIOps platform combining **traditional ML + LLM-based anomaly detection**, semantic log intelligence, and automated incident insights, with a **multi-model inference architecture** using TorchServe and FastAPI.
- Engineered **RAG pipelines with ChromaDB embeddings** for semantic search, root-cause retrieval, and structured reasoning across logs and metrics, enabling high-accuracy contextual analysis.
- Implemented **CI/CD model deployment workflows**, containerized environments, telemetry/alerts, and agent workflows for automated triage and RCA—showcasing strong experience in **production ML services, MLOps**,

distributed inference, and GenAI integration.

Enterprise RAG Chatbot

Semantic Document RAG Chatbot

- Azure, OpenAI, Embedding Ada-002, Postgres, Python, Chatbot, Streamlit
- Built a secure, multi-user platform for full-text search across large enterprise documents
- Enabled instant contextual search with highlighted matches, improving retrieval efficiency by 40%.

TECHNICAL SKILLS

Skills: AI/ML & GenAI: LLMs, RAG, Transformers, LangGraph, Prompt Engineering, Embeddings

ML Frameworks: Scikit-learn, TensorFlow (basic), Azure OpenAI, Vertex AI

Programming: Python, SQL, JavaScript, ReactJS

Cloud & Infra: Azure , GCP

APIs & Services: REST API Design, FastAPI, GitHub API, OpenAI-style inference

DevOps: Docker, Containers, CI/CD, YAML workflows, Kubernetes (beginner)

Data Engineering: ETL/ELT, Pipelines, Data Modeling, Vector Databases (Chroma)

Tools: Jupyter, Git, Postgres, MongoDB

CERTIFICATIONS

- [Google Project Management Professional Certificate](#)
- [Google Cloud Skills profile](#)
- [Microsoft Learn Profile](#)